

PROJECT INTELLIGENCE HUB

Complete Running, Testing, Synchronization and VS Code Prompt Guide

PURPOSE

This guide gives copy-ready PowerShell commands, VS Code terminal commands, sync commands, test commands, and Codex prompt templates for running and maintaining the Streamlit and Next.js versions of Project Intelligence Hub.

1. Workspace Map

Area	Path	Use
Main Streamlit workspace	C:\Users\pc\OneDrive\Documents\Project Intelligence Hub	Original Streamlit application, project templates, data folders, sync scripts and dashboard.py.
Next.js website workspace	D:\Project Intelligence Hub NextJS	Vercel-ready website copy, generated JSON data, AI routes, reports, tests and output package.
Website app	D:\Project Intelligence Hub NextJS\website	Next.js source, package.json scripts, public/data JSON, API routes and React components.
Generated reports	D:\Project Intelligence Hub NextJS\reports	Word/PDF/PPTX reports and guides generated for management and handover.
Project folders	D:\Project Intelligence Hub NextJS\projects	Project-level source folders; each project should remain isolated from other projects.
Automatic HTML outputs	D:\Project Intelligence Hub NextJS\11-outputs	One output folder per recognized project, generated HTML reports only.

2. Quick Start Commands

Need	Where to run	Command
Run Streamlit app	D:\Project Intelligence Hub NextJS	.\RUN_APP.bat
Run full no-Git sync watcher	D:\Project Intelligence Hub NextJS	.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat Watch 30
Run one-time sync now	D:\Project Intelligence Hub NextJS	.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat Once 30

generate website JSON	D:\Project Intelligence Hub NextJS\website	npm run generate-data
in Next.js locally	D:\Project Intelligence Hub NextJS\website	npm run dev
Build Next.js for Vercel	D:\Project Intelligence Hub NextJS\website	npm run build
Validate project isolation	D:\Project Intelligence Hub NextJS	python tools\validate_project_isolation.py
Run Python tests	D:\Project Intelligence Hub NextJS	python -m pytest -q tests -p no:cacheprovider

3. PowerShell Run Blocks

3.1 Run Streamlit

```
cd /d "D:\Project Intelligence Hub NextJS"
.\RUN_APP.bat
```

3.2 Run Next.js Website

```
cd /d "D:\Project Intelligence Hub NextJS\website"
npm run generate-data
npm run dev
# Open: http://localhost:3000
```

3.3 Build Website for Vercel

```
cd /d "D:\Project Intelligence Hub NextJS\website"
npm run generate-data
npm run build
```

3.4 Run Sync Once

```
cd /d "D:\Project Intelligence Hub NextJS"
```

```
.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat Once 30
```

3.5 Run Sync Watcher

```
cd /d "D:\Project Intelligence Hub NextJS"
.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat Watch 30
# Keeps checking changes every 30 seconds.
```

4. Testing and Validation Commands

Check	Command	Expected result
Compile Streamlit dashboard	python -m py_compile dashboard.py contract_claims_center.py	No syntax errors.
Compile Python services	python -m py_compile tools\generate_nextjs_website_data.py tools\validate_project_isolation.py	No syntax errors.
Run tests	python -m pytest -q tests -p no:cacheprovider	All tests pass or clear failure names are shown.
Validate isolation	python tools\validate_project_isolation.py	PASS lines for project discovery, isolation, outputs and sync files.
Generate data	cd website && npm run generate-data	portfolio.json and project JSON files update.
Build website	cd website && npm run build	Next.js production build completes.
AI health endpoint	curl http://localhost:3000/api/health/ai	Controlled status; no secret values exposed.
Unified search endpoint	curl -X POST http://localhost:3000/api/intelligence/search -H "Content-Type: application/json" -d '{"query":"show delay risks","mode":"portfolio"}'	Returns matched project intelligence or fallback answer.

5. VS Code Terminal Setup

- Open VS Code, then choose File > Open Folder and select D:\Project Intelligence Hub NextJS.
- Use Terminal > New Terminal. Confirm the prompt path is D:\Project Intelligence Hub NextJS.
- For website commands, run: cd website before npm commands.
- Keep one terminal for the app, one terminal for tests, and one terminal for sync watcher.

- Do not paste tokens into code files. Put credentials in environment variables only, such as GITHUB_TOKEN, GH_TOKEN, GROQ_API_KEY, or OPENAI_API_KEY when needed.

VS Code terminal	Use it for	Command
Terminal 1	Streamlit app	.\RUN_APP.bat
Terminal 2	Next.js website	cd website; npm run dev
Terminal 3	Tests and build	python -m pytest -q tests -p no:cacheprovider
Terminal 4	Repository sync	.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat Watch 30

6. Sync Rules

IMPORTANT

RUN_FULL_PROJECT_NO_GIT_SYNC.bat uses tools\github_no_git_sync.ps1 and reads credentials only from GITHUB_TOKEN or GH_TOKEN. Codespaces user secrets do not authenticate the local Windows process.

Task	Command	Notes
Set token for current PowerShell session	\$env:GITHUB_TOKEN='your_token_here'	Session only. Safer than writing token into files.
Set token permanently for current Windows user	setx GITHUB_TOKEN "\"your_token_here\""	Open a new terminal after this.
Run one-time push	.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat Once 30	Uploads changed/new local files according to config.
Run continuous push	.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat Watch 30	Immediate sync, then repeats every 30 seconds.
Check sync log	Get-Content .\12-logs\github_sync.log -Tail 80	Shows scanned, changed, uploaded and failed files.
Check config	Get-Content .\tools\github_sync_config.json	Target repo, excludes and deletion policy.

7. PowerShell Prompt Pack

7.1 Full Health Check Prompt

```
cd /d "D:\Project Intelligence Hub NextJS"
python -m py_compile dashboard.py contract_claims_center.py tools\generate_nextjs_website_data.py tools\validate_project_isolation.py
```

```
python tools\validate_project_isolation.py
python -m pytest -q tests -p no:cacheprovider
cd website
npm run generate-data
npm run build
```

7.2 Sync and Verify Prompt

```
cd /d "D:\Project Intelligence Hub NextJS"
.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat Once 30
Get-Content .\12-logs\github_sync.log -Tail 80
```

7.3 Website Run Prompt

```
cd /d "D:\Project Intelligence Hub NextJS\website"
npm run generate-data
npm run dev
```

8. Codex / VS Code Prompt Templates

me	Copy-ready prompt
Verify	Run the Project Intelligence Hub locally. Start Streamlit and Next.js if needed, then verify health, project dropdown, project isolation, Decision Making Dashboard, Delay Analysis, Claims Intelligence, Letters Intelligence, Output Studio, and generated outputs. Do not change code unless a failure is
Improvement	Before changing anything, inspect the relevant files and create a return point. Make one focused improvement, preserve project isolation, run tests, Next.js build, and summarize files changed and validation results.
	Run RUN_FULL_PROJECT_NO_GIT_SYNC.bat in one-time mode, then inspect 12-logs/github_sync.log and confirm code, project folders, generated data, and website files were detected for synchronization.
Isolation	Check why selected project data is mixed. Trace project_key/project_id from project discovery to dashboard, letters, claims, Delay TIA, Output Studio, and generated JSON and cache. Fix only the binding issue and run isolation validation.
Management output	Regenerate the board report/presentation from current project data only. Do not hardcode values. Use portfolio.json and project JSON as sources for Word/PDF/PPTX, and visually verify outputs.

9. Troubleshooting

Problem	Likely cause	Fix
Streamlit app does not open	Port 8755 blocked or virtual environment missing.	Run <code>.\RUN_APP.bat</code> from project root; confirm <code>.venv\Scripts\python.exe</code> exists.
Next.js build fails	Missing npm install, bad TypeScript, or generated data issue.	<code>cd website; npm install; npm run generate-data; npm run build</code>
Sync says 401 Unauthorized	GITHUB_TOKEN/GH_TOKEN missing, expired, or lacks repo access.	Set a valid token in environment variable, then open a new terminal and rerun sync.
All projects show same data	Project key/context not passed into a loader or stale cache.	Run <code>python tools\validate_project_isolation.py</code> and inspect <code>project_key</code> filters.
Mobile downloaded HTML tabs not working	JavaScript tab logic or inline assets not included in generated HTML.	Regenerate outputs and test downloaded HTML on desktop and mobile browser.
AI not working	GROQ_API_KEY missing or provider unavailable.	Set GROQ_API_KEY and verify <code>/api/health/ai</code> ; fallback should still show matched evidence.

10. Recommended Daily Operating Routine

Time	Action	Command / Check
Start of day	Run app and sync watcher	<code>.\RUN_APP.bat</code> and <code>.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat</code> Watch 30
After project file updates	Regenerate data and outputs	<code>cd website; npm run generate-data; npm run build</code>
Before sharing	Run validation	<code>python tools\validate_project_isolation.py; python -m pytest -q tests -p no:cacheprovider</code>
Before board review	Open Decision Making Dashboard and generated HTML reports	Confirm figures, warnings, and decisions are current.
End of day	One-time sync and log review	<code>.\RUN_FULL_PROJECT_NO_GIT_SYNC.bat</code> Once 30; <code>Get-Content .\12-logs\github_sync.log -Tail 80</code>

11. Command Ownership

- PowerShell runs the application, validation, sync and build commands.
- VS Code is the working environment for editing, launching terminals and keeping prompts saved.
- Codex prompts should always mention: preserve data pipeline, preserve project isolation, run validation, and do not hardcode values.
- GitHub/Vercel alignment depends on successful sync and a successful website build.

FINAL RULE

When in doubt, run the sequence: generate data -> validate project isolation -> run tests -> build website -> run one-time sync -> inspect sync log. This protects project isolation and keeps Streamlit, Next.js, GitHub and Vercel aligned.